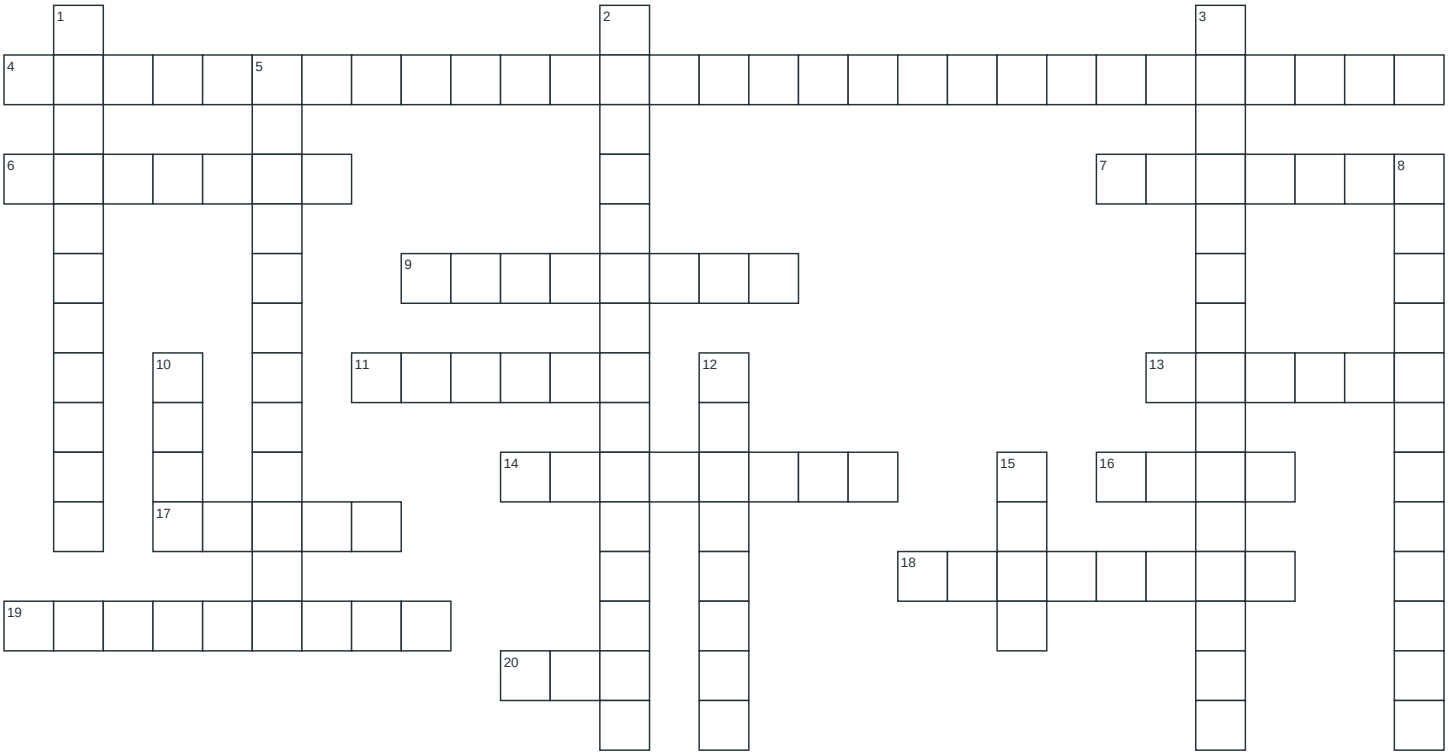
Momentum and Collisions Crossword

Hard | 20 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



ACROSS

4. Characterizes a perfectly inelastic collision
6. The change in momentum caused by a force over time
7. Resistance to changes in motion
9. Mass multiplied by velocity
11. The gravitational force acting on an object
13. Backward motion associated with another object's forward momentum
14. Force that opposes relative motion between surfaces in contact.
16. Measure of the amount of matter in an object.
17. Quantity that is a scalar
18. The total length of a traveled path
19. Orientation of motion, needed to describe a velocity vector.

ACROSS (continued)

20. Relationship that expresses Newton's second law for constant mass

DOWN

1. Force that is a surface's push perpendicular to that surface
2. Forces between interacting objects are equal in magnitude and opposite in direction.
3. SI-derived unit commonly used to measure speed or velocity.
5. The directed change from initial to final position
8. Change in velocity per unit time.
10. Unit that can express momentum
12. The vector sum of all forces acting on an object
15. State in which position does not change relative to a chosen reference point.

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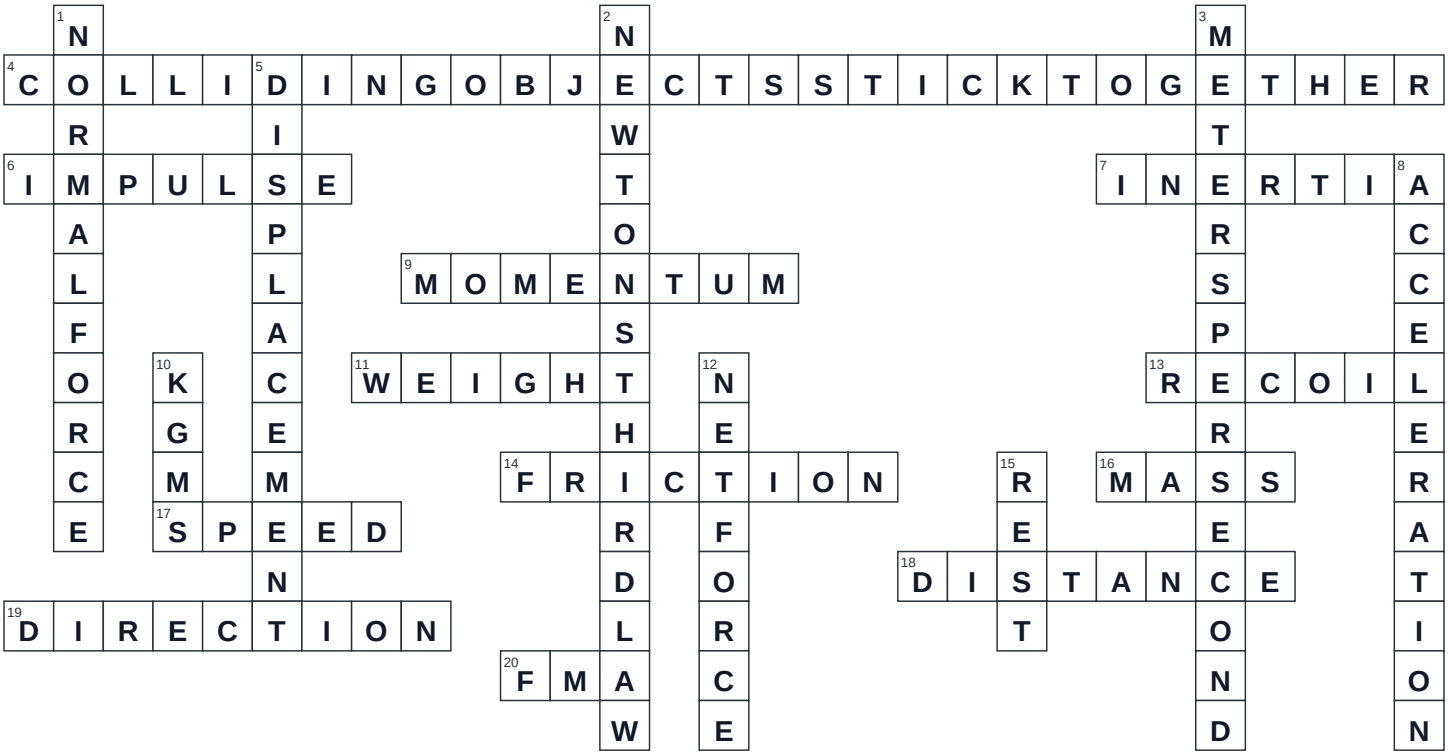
Hard | 20 clues | Teacher Answer Key

TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.



ACROSS

4. Colliding Objects Stick Together
6. Impulse
7. Inertia
9. Momentum
11. Weight
13. Recoil
14. Friction
16. Mass
17. Speed
18. Distance
19. Direction

ACROSS (continued)

20. F = ma

DOWN

1. Normal Force
2. Newton’s third law
3. Meters per second
5. Displacement
8. Acceleration
10. Kg·m/s
12. Net Force
15. Rest

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